Literature Status: A Complex $n_L(X) < n(X)$ Example

FA/Banach run `fa_banach_001`

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Status

This is a literature-already-answered packet, not an original proof from the run. Wang–Huang–Tan, arXiv:1211.5753, ask at the end of Section 4 whether there exists a Banach space X such that

$$n_L(X) < n(X).$$

Perez-Hernandez, arXiv:2507.18208, answers the broad real-or-complex question affirmatively in the complex case.

Original Problem

The original source is:

> R. Wang, X. Huang, D. Tan, *On the numerical radius of Lipschitz operators in Banach spaces*, arXiv:1211.5753.

The paper defines $\mathrm{Lip}_0(X)$ as all Lipschitz maps $X \to X$ which vanish at 0, and defines $n_L(X)$ by taking the infimum of the Lipschitz numerical radius over norm-one maps in this class. The final problem asks whether strict inequality $n_L(X) < n(X)$ can happen.

Later Answer

The supporting source is:

> A. Perez-Hernandez, *Lipschitz vs Linear Numerical Index in certain Banach spaces*, arXiv:2507.18208.

In its introduction, the supporting source says that the answer depends on the underlying scalar field. For the complex Hilbert plane

$$X = \mathbb{C}^2, \qquad \|(z_1, z_2)\| = (|z_1|^2 + |z_2|^2)^{1/2},$$

the classical numerical index is $n(\mathbb{C}^2) = 1/2$. The paper then considers the real-linear Lipschitz map

$$F(z_1, z_2) = (\overline{z_2}, -\overline{z_1}).$$

It has $\|F\|_L = 1$. If $y = (w_1, w_2) \in S_X$, the unique norming complex-linear functional is

$$x^*(z_1, z_2) = \overline{w_1} z_1 + \overline{w_2} z_2.$$

Therefore

$$x^*(F(y)) = \overline{w_1 w_2} - \overline{w_2 w_1} = 0.$$

Thus the Lipschitz numerical range of F is $\{0\}$, so

$$n_L(\mathbb{C}^2) = 0 < \frac{1}{2} = n(\mathbb{C}^2).$$

Verification Notes

The 2025 definition of $W_L(F)$ is equivalent to the 2012 definition: a unit functional norming $x_1 - x_2$ is the normalized version of an element of $D(x_1 - x_2)$. The map F is admissible because the Lipschitz numerical index is taken over all Lipschitz maps fixing the origin, not only over complex-linear maps.

The 2025 paper keeps the real case open in full generality and proves equality for real separable spaces and real dual spaces. This packet records only the complex affirmative answer to the original broad question.

Search Bounds

The local registry, solution index, attempt index, and proof-gap index were searched for arXiv:1211.5753 and core phrases such as $n_L(X) < n(X)$, Lipschitz numerical index, and numerical radius. No exact prior packet was found. The decisive later source passage was found in the local parsed source for arXiv:2507.18208, whose bibliography identifies arXiv:1211.5753 as the Wang–Huang–Tan paper.